BULGARIAN NATIONAL OLYMPIAD IN MATHEMATICS

Fourth round

1995

Problem 1. Find the number of all integers $n > 1$, for which the number $a^{25} - a$ is divisible by n for every integer a.

Problem 2. A triangle ABC with semiperimeter p is given. Points E and F lie on the line AB and $CE = CF = p$. Prove that the excircle k_1 of $\triangle ABC$ to the side AB touches the circumcircle k of $\triangle EFC$.

Problem 3. Two players **A** and **B** take stones one after the other from a heap with $n \geq 2$ stones. **A** begins the game and takes at least 1 stone but no more then $n - 1$ stones. Each player on his turn must take at least 1 stone but no more than the other player has taken before him. The player who takes the last stone is the winner. Find who of the players has a winning strategy.

Problem 4. The points C_1, A_1 and B_1 lie on the sides AB, BC and CA of the equilateral triangle ABC respectively in such a way that the inradii of the triangles C_1AB_1, B_1CA_1, A_1BC_1 and $A_1B_1C_1$ are equal. Prove that A_1, B_1 and C_1 are the midpoints of the corresponding sides.

Problem 5. Let $A = \{1, 2, \ldots, m + n\}$, where m and n are positive integers and let the function $f : A \to A$ be defined by the equations:

$$f(i) = i + 1 \quad \text{for} \quad i = 1, 2, \ldots, m - 1, m + 1, \ldots, m + n - 1$$

$$f(m) = 1 \quad \text{and} \quad f(m + n) = m + 1.$$

a) Prove that if m and n are odd then there exists a function $g : A \to A$ such that $g(g(a)) = f(a)$ for all $a \in A$.

b) Prove that if m is even then $m = n$ iff there exists a function $g : A \to A$ such that $g(g(a)) = f(a)$ for all $a \in A$.

Problem 6. Let x and y be different real numbers such that $\dfrac{x^n - y^n}{x - y}$ is an integer for some four consecutive positive integers n. Prove that $\dfrac{x^n - y^n}{x - y}$ is integer for all positive integers n.

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